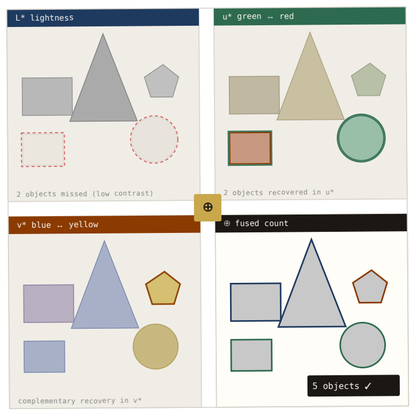


Multi-Channel LUV Approach for Robust Object Counting using Raspberry Pi-based Computer Vision

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1. INTRODUCTION

Object counting is a core image-processing task used in industrial inspection, traffic monitoring, and more. It is achievable with just a Raspberry Pi, a camera, and Python/OpenCV [1].

Existing methods convert images to grayscale, which works under sharp contrast but fails when shadows or varying camera angles reduce visibility [1].

By switching to the LUV color space, we decouple lightness (L) from chromaticity (u, v), enabling more robust object detection where grayscale falls short [2].



Figure 1.1: Objects with low contrast and close distance [1]

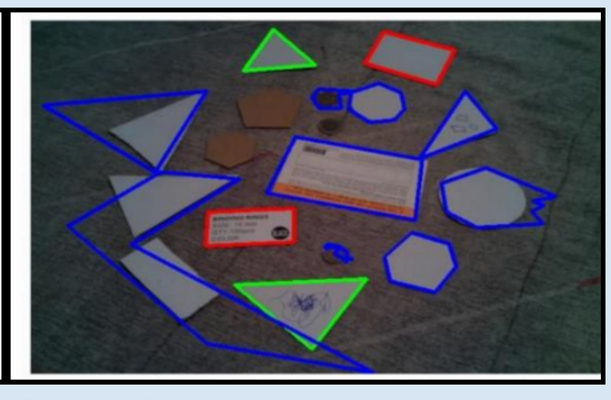


Figure 1.2: Object detection result with gray scale filtering [1]

2. RESEARCH QUESTION

Can counting accuracy under low-contrast and close-spacing conditions be improved by replacing grayscale with the perceptually-uniform CIE L*u*v* color space — processing the L*, u*, and v* channels separately and fusing the results?

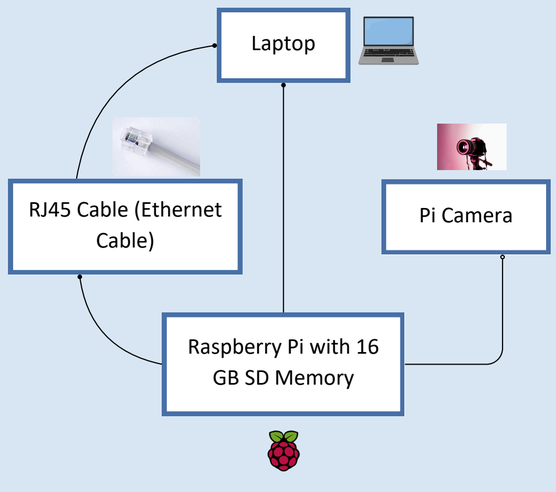


Figure 2: Hardware Design Procedure [1]

Test Scenario	Objects	Baseline Accuracy
Mixed shapes	13	92.31%
Poor spacing / weak contrast	17	58.72%
Shadow & close-up	—	errors

Table 1: Baseline Accuracy [1]

3. METHODOLOGY

RGB channels range from 0–255, while CIE L*u*v* separates lightness (L*: 0–100) from chromaticity (u*: −134 to +220, v*: −140 to +122).

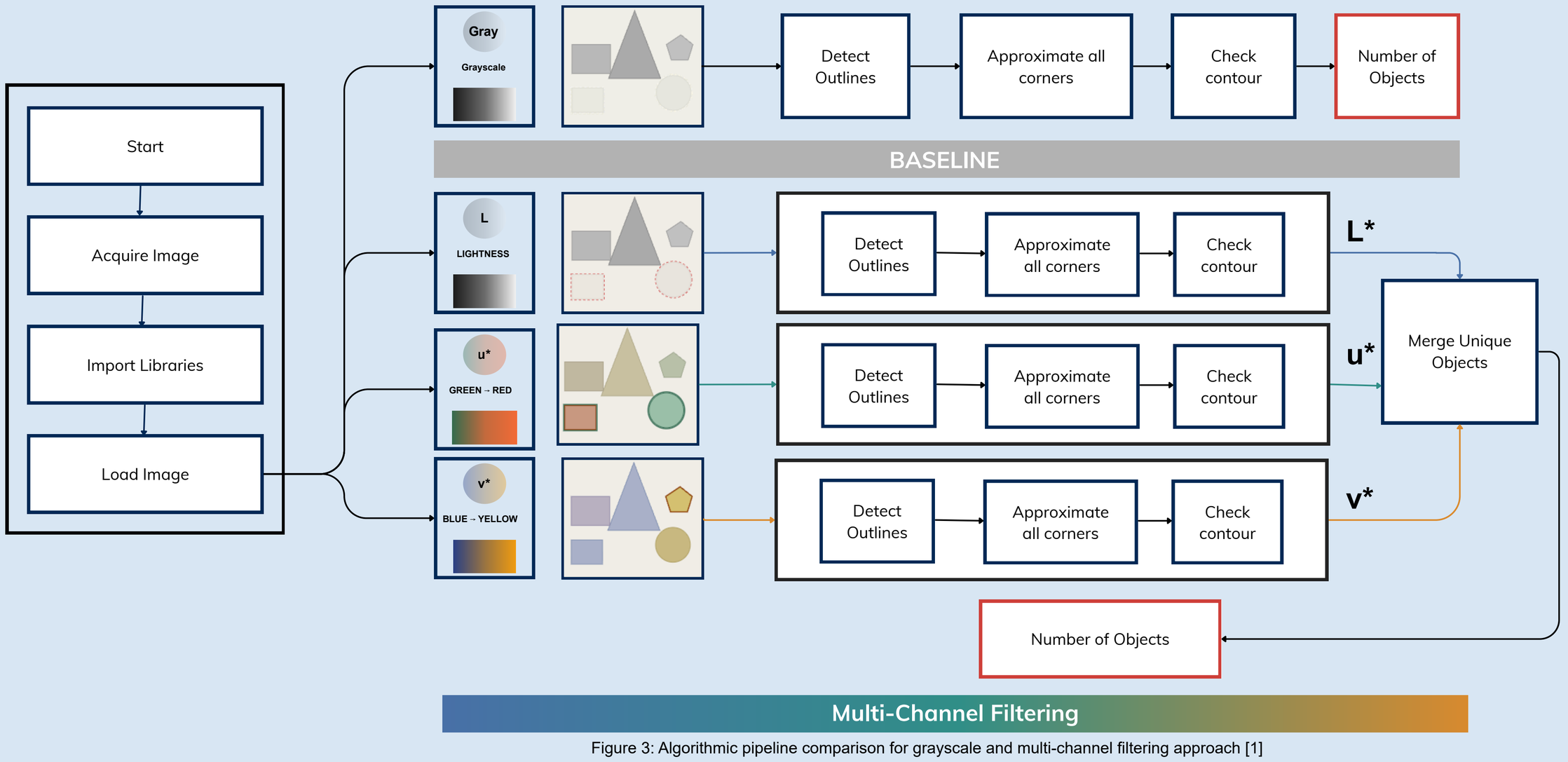


Figure 3: Algorithmic pipeline comparison for grayscale and multi-channel filtering approach [1]

References :

[1] Abdulhamid, M., Odondi, O., & Al-Rawi, M. (2020). Computer Vision Based on Raspberry Pi System. Applied Computer Science, 16(4), 85–102. doi:10.23743/acs-2020-31.

[2] Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson.

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